



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. EEE

Course Code & Title: EE3601 Protection and Switchgear

Name of the Faculty member (s): Mr. E. Thangam

Innovative Practice Description

- **Unit / Topic:** Unit I / Essential Qualities of Protection

- **Course Outcome:** CO1

- **Topic Learning Outcome:** TLO1

- **Activity Chosen:** One-minute paper

- **Justification:**

Since qualities of protection is an important topic in concept wise and university exam point of view, I have chosen this topic and the reason for choosing One-minute paper activity is while recollecting the concepts and hints learnt in the classroom, the students will be able to reproduce the concept in the exam.

- **Time Allotted for the Activity:** 03 Minutes

- **Details of the Implementation:**

(A brief content about how the practice was followed)

After teaching the concepts, I asked the students to think about the topic without writing anything for 2 minutes.

Total strength: 60

Reporter: Myself

At the end of the class (Last 4 minutes)

- ✓ I asked the students to think about essential qualities of protection, and how it is influence on the equipment protection? For 2 minutes
- ✓ Then I asked the students to write as much as they can within a short period of time(1 minute)
- ✓ Finally, I collected the papers form each students (1 Minute)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO6	PO7	PO8	PO9	PO10	PO12	PSO1	PSO2
CO1	3	2	1	1	1	1	1	2	1	1	1

(1 – Low

2 – Moderate

3 – High)



- PO / PSO mapped:

Innovative practice	PO10
	2
Justification for correlation	Due the One minute paper activity student's written communication will be improved. So it is mapped moderately

- Images / Screenshot of the practice:





1. Sensitivity:

It is the ability of a protective system to choose the faulty part of the system alone and disconnect that part without affecting the rest of the healthy part of the system.

2. Fast operation:

It is defined as the ability of a protective system to choose and disconnect the faulty part as quickly as possible.

3. Selectivity:

It is the selectivity of the relay system to operate with even low attracting quantity.

4. Reliability:

It is the measure of how reliable a protective system is to call upon to disconnect the faulty part.

5. Stability:

It is the ability of a protective system to remain stability even fault current interruption and disturbance due to them. The protective system should not be changed due to the any fault current etc.

• Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

Students were asked me to conduct such activities for forth coming classes.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt good when written most of the points by students.
- ✓ While conducting the activity, I understood that the students will be able to explain the essential qualities of protection.

❖ **Challenges faced in implementation:**

- ✓ Time utilization for this activity
- ✓ Slow learners were not able reproduce the much points.

References:

- ❖ Sunil S.Rao, 'Switchgear and Protection', Khanna Publishers, New Delhi, 2008.
- ❖ Badri Ram, B.H. Vishwakarma, 'Power System Protection and Switchgear', New Age International Pvt. Ltd Publishers, Second Edition 2011.

Signature of Faculty Member

HOD



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Innovative Practice Description

- **Unit / Topic:** Unit II / Introduction, Operating Principle of Relays.
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** Visible Quiz
- **Justification:**

This Visible Quiz on the topic "Introduction and Operating Principle of Relays" is designed to actively engage students in reinforcing foundational concepts related to protective relays used in electrical engineering. By incorporating a visual and interactive format, the quiz enhances conceptual clarity, encourages real-time application of theoretical knowledge, and promotes active recall, which aids in better retention. This activity also serves as a formative assessment tool, helping instructors identify areas where students may need further clarification, thereby improving the overall learning outcomes.

- **Time Allotted for the Activity:** 8 Minutes
- **Details of the Implementation:**

The Visible Quiz on "Introduction and Operating Principle of Relays" was conducted as an in-class interactive activity using a projector and visual aids such as relay diagrams, animations, and circuit simulations. Questions were displayed one at a time, allowing students to visualize the concepts before selecting the correct answer. Students participated individually or in small groups, submitting their responses through printed answer sheets or digital platforms (e.g., Google Forms or learning management systems). The quiz covered key aspects like types of relays, their construction, working principles, and applications. Immediate feedback and brief discussions were facilitated after each question to reinforce learning. The entire activity was completed within a 8 minute session.

Total strength: 63

Reporter: Myself

At the end of the class (Last 4 minutes)

- ✓ I asked the students to think about essential qualities of protection, and how it is influence on the equipment protection? For 2 minutes



- ✓ Then I asked the students to write as much as they can within a short period of time(1 minute)
- ✓ Finally, I collected the papers form each students (1 Minute)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO6	PO7	PO8	PO9	PO10	PO12	PSO1	PSO2
CO2	3	2	2	1	1	1	1	1	2	2	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10
	2
Justification for correlation	Due the visible quiz activity student's oral communication will be improved. So it is mapped moderately

- **Images / Screenshot of the practice:**





- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students were asked me to conduct such activities for forth coming classes.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt good when written most of the points by students.
 - ✓ While conducting the activity, I understood that the students will be able to explain the essential qualities of protection.

- ❖ ***Challenges faced in implementation:***

- ✓ Time utilization for this activity
 - ✓ Slow learners were not able reproduce the much points.

References:

- ❖ Sunil S.Rao, 'Switchgear and Protection', Khanna Publishers, New Delhi, 2008.
- ❖ Badri Ram, B.H. Vishwakarma, 'Power System Protection and Switchgear', New Age International Pvt. Ltd Publishers, Second Edition 2011.

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Innovative Practice Description

- **Unit / Topic:** Unit III / Current Transformer and their applications in protection schemes
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO - 10
- **Activity Chosen:** Demonstration

- **Justification:**

The Demonstration on "Current Transformer (CT) and their Applications in Protection Schemes" is aimed at providing students with a practical understanding of how CTs function within electrical power systems. This activity bridges the gap between theoretical knowledge and real-world applications by visually showcasing the construction, working, and significance of CTs in measuring and protecting high-voltage circuits. Through hands-on observation or simulation-based demonstration, students grasp how CTs enable safe current measurement and trigger protective devices during faults. This experiential learning approach enhances conceptual clarity, encourages inquiry, and strengthens students' ability to analyze and apply protection strategies effectively in engineering practice.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

The Demonstration on "Current Transformer and Their Applications in Protection Schemes" was conducted in a laboratory or classroom setting equipped with electrical training kits or simulation software. A functional CT setup was used to show its construction, working principle, and connection with protective relays. The instructor explained how CTs step down high currents for measurement and how they interact with relays during fault conditions. Students observed live readings, fault simulations, and relay operations triggered by the CT output. Safety precautions were emphasized throughout the session. Alternatively, where physical setup was not available, software tools like MATLAB/Simulink or virtual labs were used to demonstrate CT behavior in protection schemes. Students were encouraged to ask questions and discuss practical use cases, making the session interactive and application-focused.

After teaching the concept, give students one or two minutes to think about the topic without writing anything.



Total

Strength is 63,

Photographer: one student – Mr. Baskar (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about CT connections.
- ✓ Then I told them to Pair with their neighbours and discuss about the operation of CT for another 1 minute.
- ✓ Finally, I shown the various CT to each students.

• **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 5	PO 10	PO12	PSO 1
CO3	3	2	1	2	2	2
(1 – Low 2 – Moderate 3 – High)						

• **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	Current Transformer and their applications in protection schemes. So it is correlated slightly.

• **Images / Screenshot of the practice:**





Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students told that it is good see all the parts of the CT individually and demonstration helps the students to understand the concepts easily.

❖ *Benefit of the practice:*

1. Students can able to understand the impact of engineering solution on society
2. Most of the students attended the question asked Internal Assessment Test – I retest.
3. The success of the activity was evaluated by asking the same question during IAT-I. **Around 80% of students answered.**
4. Students can able to explain the concepts in examination without any confusion.

❖ *Challenges faced in implementation: Nil*

References:

- ❖ Sunil S.Rao, 'Switchgear and Protection', Khanna Publishers, New Delhi, 2008.
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Innovative Practice Description

- **Unit / Topic:** Unit V / Vacuum Circuit Breakers

- **Course Outcome:** CO5

- **Topic Learning Outcome:** TLO12

- **Activity Chosen:** Field Visit

- **Justification:**

The topic circuit breaker has different types, since each type has used for different voltage level and applications. After teaching the constructional, I thought of conducting this field visit for making the students to give the input about vacuum circuit breaker which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 45 Minutes

- **Details of the Implementation:**

After teaching the concept, students were taken to the RIT – Power House Switch gear yard and ask them to think about the constructional features of circuit breakers.

Total Strength is 64,

Photographer: one student - Mr. Gowtham Guru (interested in photography)

Reporter: Myself

At the end the visit (Last 5 minutes)

- ✓ I asked the students to think about various parts of circuit breaker for 2 minutes.
- ✓ Then I told them to Pair with their neighbors and discuss about the construction and operation of circuit breaker for another 1 minute.
- ✓ Finally, I randomly asked students about the construction parts and operation of circuit breaker.(2 minutes)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10	PO12	PSO1
CO1	3	2	1	1	1	2	1
(1 – Low			2 – Moderate		3 – High)		



- PO / PSO mapped:

Innovative practice	PO7
	1
Justification for correlation	Due to this activity student's will understand the impact of technical solution on environment. So it is slightly correlated.

- Images / Screenshot of the practice:



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were asked me to conduct such activities for maximum topics.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt good when I am asking the question most of the points shared by students.
- ✓ While conducting the activity, I understood that the students will be able to recollect the points about circuit breakers.

❖ *Challenges faced in implementation:*



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✓ Time utilization for this activity

References:

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